

LEVEL 1 — FOUNDATIONS & ENTRY-LEVEL DEVOPS

Target Roles

- Junior DevOps Engineer
 - Cloud Support Associate
 - Cloud Operations Engineer
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Skills To Learn

Linux & Shell

- [Learn Linux - The Full Course](#)
- [60 Linux Commands you NEED to know \(in 10 minutes\)](#)
- [Every LINUX Concept Explained in 8 Minutes.](#)

Git & GitHub

- [Git & GitHub Crash Course for Beginners \[2026\]](#)

Networking Fundamentals

Topics:

- DNS
- HTTP vs HTTPS
- REST APIs
- Load Balancing
- Reverse Proxy
- SSL/TLS

Resources:

- [Learn Networking In 25 MINUTES \[Crash Course\] | Networking Fundamentals ...](#)

Cloud Fundamentals

Topics:

- Compute
- Storage
- IAM
- Networking

Resources:

- [Cloud Computing Explained: The Most Important Concepts To Know](#)

Notes:

Scripting Basics\

Bash

- [Bash Scripting Tutorial for Beginners](#)

Intro To AI Systems

Topics:

- AI APIs
- Prompt flow
- High-level AI architecture

Resources:

- [\[1hr Talk\] Intro to Large Language Models](#)
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Certifications (Optional)

- AWS Cloud Practitioner
- GCP Digital Leader

Resources:

- [AWS Certified Cloud Practitioner Certification Course \(CLF-C02\) - Pass the Ex...](#)
 - [Google Cloud Digital Leader Certification Course - Pass the Exam!](#)
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LEVEL 2 — CLOUD & AUTOMATION ENGINEER

Target Roles

- DevOps Engineer
 - Cloud Engineer
 - Infrastructure Engineer
 - Build & Release Engineer
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Skills To Learn

CI/CD

GitHub Actions

- [Complete GitHub Actions Course - From BEGINNER to PRO](#)

Jenkins

- [Jenkins Tutorial For Beginners | Jenkins Full Course | Jenkins Tutorial](#)

Infrastructure as Code

Terraform

- [Complete Terraform Course - From BEGINNER to PRO! \(Learn Infrastructure a...](#)

Pulumi

- [Pulumi Crash Course](#)

Docker

- <https://www.tech5ense.com/p/cloud-devops-roadmap-phase-2>

System Design Basics

Topics:

- Scalability
- Reliability
- Caching

- CDN
- Load Balancing

Resources:

- [System Design Course – APIs, Databases, Caching, CDNs, Load Balancing & ...](#)
- [System Design Concepts CRASH COURSE For Beginners | End-To-End DevO...](#)

Intro To AI Infrastructure

Topics:

- GPU basics
- AI workloads
- AI deployment fundamentals

Resources:

- [NVIDIA-Certified Associate AI Infrastructure and Operations \(NCA AIIO\) Free S...](#)
 - Certification:
<https://www.nvidia.com/en-us/learn/certification/ai-infrastructure-operations-associate/>
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Certifications (Optional)

- Terraform Associate
- AWS DevOps Engineer
- Azure DevOps Engineer

Resources:

- [HashiCorp Terraform Associate Certification Course \(003\) - Pass the Exam!](#)
 - [AWS Certified DevOps Engineer Professional](#)
 - [Azure DevOps Engineer Expert Certification \(AZ-400\) – Full Course to PASS th...](#)
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LEVEL 3 — KUBERNETES & CLOUD NATIVE ENGINEERING

Target Roles

- Senior DevOps Engineer

- Kubernetes Engineer
 - Platform Engineer
 - Cloud Native Engineer
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Skills To Learn

Kubernetes Fundamentals

Topics:

- Pods
- Deployments
- Services
- Ingress

Advanced Kubernetes

Topics:

- HPA
- VPA
- KEDA
- Operators
- Persistent Volumes

Resources:

- [Complete Kubernetes Tutorial for Beginners](#)
-  Kubernetes Autoscaling: HPA vs. VPA vs. Keda vs. CA vs. Karpenter vs. Fargate

Helm

-  You will never forget HELM after watching this.

Service Mesh

Istio

-  Istio Crash Course: Complete Guide to Service Mesh & Certified Associate Exa...

Linkerd

-  Introduction to Linkerd for beginners | a Service Mesh

Observability

Topics:

- Prometheus
- Grafana
- OpenTelemetry

Resources:

-  Learn Observability in 5 hours | Tool wise Demo + Complete Demo using Open...

AI Workloads On Kubernetes

Topics:

- GPU scheduling
- Batch inference
- Real-time inference

Resources:

-  GPUs in Kubernetes for AI Workloads
-  Building Inference-as-a-Service on Kubernetes
-  AI Inference: The Secret to AI's Superpowers

Certifications (Optional)

- CKA
- CKAD
- CKS

Resources:

- <https://www.tech5ense.com/p/your-kubernetes-certs-playbook>

LEVEL 4 — PLATFORM & GITOPS ENGINEERING

Target Roles

- Staff DevOps Engineer
 - GitOps Engineer
 - Infrastructure Architect
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Skills To Learn

GitOps

ArgoCD

-  ArgoCD Tutorial for Beginners | GitOps CD for Kubernetes

FluxCD

- [Master FluxCD GitOps: Manage Kubernetes Clusters Declaratively](#)

Multi-Cluster Architecture

-  Multi Cluster Deployment with GitOps | Step by Step Guide | DevOps Project | ...

Security & Policy Enforcement

OPA Gatekeeper

-  Secure Your Kubernetes Clusters with OPA Gatekeeper: Policy and Governanc...

Deployment Strategies

Topics:

- Canary Deployments
- Blue-Green Deployments
- Rolling Updates

Resources:

-  Top 5 Most-Used Deployment Strategies

Cost Optimization

- [Cloud Cost Optimization Strategy Explained | FinOps](#)
- [Day-18 | AWS Cost Optimization | Most Popular Cloud and DevOps project| Ev...](#)

AI Platform Integration

Topics:

- AI deployment workflows
- Training vs inference infrastructure

Resources:

- [AI Engineering in 76 Minutes \(Complete Course/Speedrun!\)](#)
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LEVEL 5 — PLATFORM ENGINEERING & SRE

Target Roles

- Staff SRE
 - Platform Engineering Lead
 - Principal DevOps Engineer
 - Solutions Architect
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Skills To Learn

Internal Developer Platforms

Backstage

- [Getting Started with Backstage: From Zero to Operational Dev Portal](#)

Crossplane

- [Complete Crossplane Playlist](#)

Reliability Engineering

Topics:

- SLO
- SLA
- SLI
- Error Budgets

Resources:

-  [What is SRE | Tasks and Responsibilities of an SRE | SRE vs DevOps](#)
-  [SLA vs SLO vs SLI | SRE Interview Questions | DevOps FAQ |#devopsintervie...](#)

Incident Management

- [What is Incident Management?](#)

Chaos Engineering

-  [Understanding Chaos Engineering](#)

Capacity Planning

- [What is capacity planning?](#)

AIOps

Topics:

- AI-assisted debugging
- Alert triaging
- Automated remediation

Resources:

-  [What is AIOps?](#)

Projects

Project 1

- [End-To-End DevOps + AIOps Project | COMPLETE Series](#)
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LEVEL 6 — AI DEPLOYMENT ENGINEERING

Target Roles

- Applied AI Engineer
 - AI Solutions Architect
 - Forward Deployed Engineer
 - Principal Architect
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Skills To Learn

AI Infrastructure

Topics:

- Model Serving
- RAG Systems
- Vector Databases
- AI APIs

Resources:

- <https://learn.microsoft.com/en-us/azure/databricks/machine-learning/model-serving/>
- <https://www.ibm.com/think/topics/retrieval-augmented-generation>
- <https://www.ibm.com/think/topics/vector-database>
-  MCP vs API: Simplifying AI Agent Integration with External Data

Inference Engines

vLLM

-  Understanding vLLM with a Hands On Demo
-  How the VLLM inference engine works?

TensorRT

-  Getting Started with NVIDIA Torch-TensorRT
-  TensorRT Overview

SGLang

- <https://huggingface.co/blog/paresh2806/sglang-efficient-llm-workflows>

-  Open Source Model Performance Optimization With SGLang - Yineng Zhang, ...

Agentic Workflows

Topics:

- MCP
- Tool Calling
- AI Agents

Resources:

-  What is Tool Calling? Connecting LLMs to Your Data
-  What is MCP? Integrate AI Agents with Databases & APIs
-  MCP vs API: Simplifying AI Agent Integration with External Data

AI Deployment Patterns

Topics:

- Batch inference
- Real-time inference
- Distributed inference

AI Infrastructure Optimization

Topics:

- GPU utilization
- Scaling inference systems
- Latency optimization

Resources:

-  NVIDIA-Certified Associate AI Infrastructure and Operations (NCA AIIO) Free S...
- Certification:
<https://www.nvidia.com/en-us/learn/certification/ai-infrastructure-operations-associate/>

Certifications (Optional)

- [Databricks Generative AI](#)
- [NVIDIA AI Certifications](#)

LEVEL 7 — LEADERSHIP & INFRASTRUCTURE STRATEGY

Target Roles

- Head of Platform Engineering
 - VP of Engineering
 - Director of Infrastructure
 - Distinguished Engineer
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Skills To Learn

Engineering Leadership

Topics:

- Team management
- Hiring
- Mentorship
- Technical strategy

Resource:

- [Engineering Leadership - Roles & Responsibilities | The GitLab Handbook](#)

Infrastructure Strategy

Topics:

- Multi-region architecture
- Cost governance
- Reliability trade-offs

Resources:

-  [AWS re:Invent 2024 - Best practices for creating multi-Region architectures on ...](#)
- [What is Cloud Cost Governance? | Harness Blog](#)
- [Reliability tradeoffs - Microsoft Azure Well-Architected Framework](#)

AI Governance

Topics:

- AI privacy
- Compliance
- Security
- Governance frameworks

Resource:

- [What is AI Governance? | IBM](#)

Business Alignment

Topics:

- Infrastructure ROI
- Scaling strategy
- Organizational planning

Resources:

- [What Is Cloud Scalability? | Microsoft Azure](#)
- [What Is Cloud Scalability? | Microsoft Azure](#)
- [Organization design - AWS Cloud Adoption Framework: People Perspective](#)

CERTIFICATION ROADMAP

Level	Certifications
Level 1	AWS Cloud Practitioner / GCP Digital Leader
Level 2	Terraform Associate / DevOps Engineer
Level 3	CKA / CKAD / CKS
Level 4	GitOps Associate
Level 5	Cloud Professional + NVIDIA AI Infra
Level 6	Databricks GenAI / NVIDIA AI
Level 7	-

FOR PROJECT IDEAS

- <https://github.com/NotHarshhaa/DevOps-Projects>